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Three Decades of Time Lapse Saturation Monitoring in a Mature Carbonate Field with Water and Injection Gas Injection: A Practical Workflow for Quad Phase Logs Interpretation and Data Integration

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Abstract

This paper presents insights from 30 years of time-lapse fluid saturation monitoring in a giant carbonate reservoir, Onshore Abu Dhabi. It uses multi-vintage open hole and cased hole data to track quad-phase saturation (oil, water, injection gas, reservoir gas). The flood front evolution, driven by water/injection gas injection since 1980's, was monitored using open-hole and time-lapse cased-hole pulsed neutron measurements. Given the reservoir's heterogeneous rock quality (porosity: 0-35%, permeability: 0.01-3000 mD, varying fluid properties), reliable interpretation is crucial to manage data acquired across multiple decades, different borehole environments, service providers, and tool technologies. This paper will cover two parts: Logs reinterpretation and utilization of time-lapse integrated data analysis for reservoir field development.

Quad phase neutron modelling workflow was introduced for interpretation. To solve issues with multi-vintages, cased hole porosity was recalculated from neutron ratio curves and normalized to open-hole baselines. The ratio curves, particularly near-far, showed distinction between the two gases due to density/hydrogen index difference, which has not yet equilibrated in the reservoir. This enables proper characterization due to large dynamic range. The four-fluid end points were built from compositional PVT data has substantially improved gas saturation accuracy and resolving injection gas versus hydrocarbon gas ambiguities. The model is conducted in two parts: below GOC (oil, water, reservoir gas) and above GOC (injection gas, reservoir gas, water), before recombined. By emphasizing saturation change rather than absolute values, the workflow neutralized systematic errors from different data vintages while preserving true reservoir dynamics.

Uncertainties are systematically evaluated covering data quality, tool uncertainties and borehole environment. In environment affected by mud invasion such as Logging-While-Tripping, significant gas suppression was observed while time-lapse PNL data revealed flowback, clearer GOC, and residual invasion. So, for water saturation baseline, open hole data and saturation height function were used, leaving uncertainty on gas/oil. Wells affected by invasion were flagged for conditional interpretation, ensuring GOC observations remained reliable, while saturation values were treated with caution. Representative

case studies illustrate how the workflow resolves ambiguities in complex geological settings and under varying injection schemes. In the second part of the study, dynamic saturation trends are used to refine sweep efficiency and recovery strategies. Preferential flow and non-uniform displacement patterns informed updates to geological models and reservoir management plans. Integration of time-lapse saturation with core-calibrated profiles delineated effective flow units, identified key rock types, and validated dynamic models through history matching.

This long-term study demonstrates the value of combining open and cased-hole data of multi vintages, neutron modeling, and uncertainty-aware workflows to unlock deeper understanding of reservoir architectures. Results from this study have enhanced geological characterization and improved field development strategies to increase recovery in a mature, complex carbonate field.

Geological Background

Reservoir ‘A’ is a carbonate unit within a multi-stack reservoir system located in Field A, onshore of Abu Dhabi. This field contains substantial oil and gas reserves and has been in production since the 1960s, with thousands of wells drilled to date. The 160ft thick reservoir formed in the early Cretaceous, is crossed by the regional North 75 West fault, and was deposited during a third-order sea level sequence. The depositional setting is a wide, shallow marine carbonate ramp with NE progradation of the platform, featuring various facies such as wackestone, packstone, peloidal grainstone, algal skeletal, coated grain, floatstone, and rudstones. The reservoir is oil-bearing with gas cap, consisting of nine sub-reservoir units intercalated with eight dense layers which because of the contrast are referred to as Upper and Lower zones. The structure map, depositional setting and typical logs response showing facies between Upper and Lower are illustrated in Fig. 1.

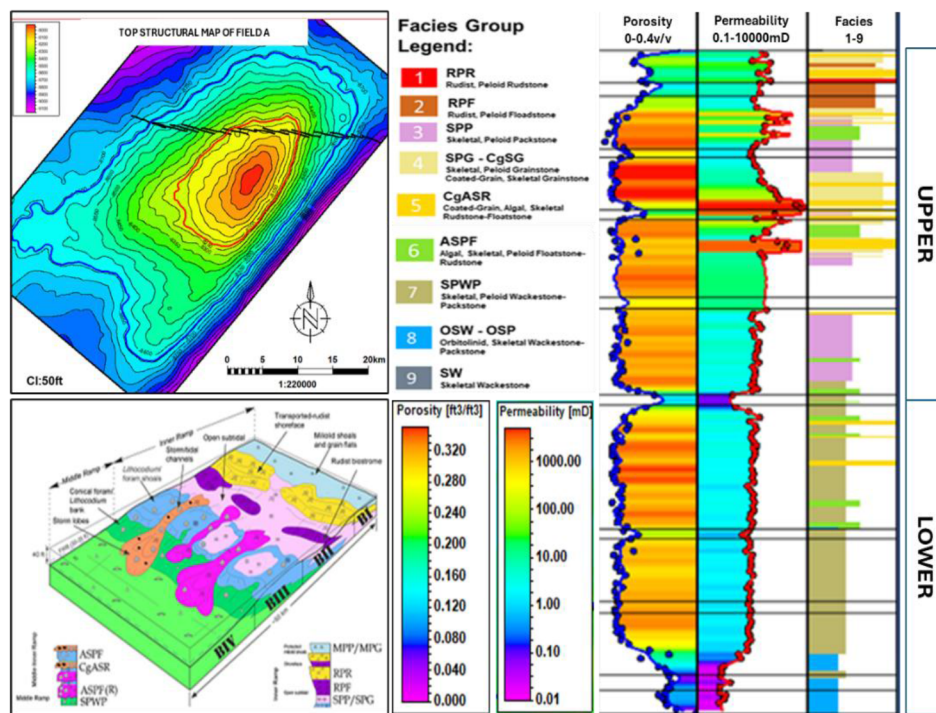


Figure 1—Structure map, depositional setting and typical logs response

Generally, both the Upper and Lower zones have very similar pore space as depicted from porosity logs averaging around 20-23pu but significant changes in permeability. The Lower reservoir zone is largely homogeneous, consisting of transgressive system tract mud-supported open marine deposits that shallow

upward to grain-dominated mid-ramp facies. The Lower zone was dominated by muddy facies such as Skeletal, Peloid Wackestone-Packestone (SPWP) and Orbitolina Skeletal Wackestone-Packestone (OSP-OSW) as products of early stage of High stand sequence. This results in consistent permeabilities within the Lower zone, ranging between 0.01 mD and 10 mD.

In contrast, the Upper reservoir zone is a high stand system tract with shallow shoal grain-dominated deposits. The shallowing upward depositional sequence extends to lagoon back-shoal muddy facies at the top of the zone. Over most of the field, leaching enhances the reservoir quality in both zones, with a more pronounced effect in the Upper zone due to the high permeability storm deposit streaks. The permeability averages around 30 mD but ranges from 1 mD to 3000 mD. These mainly contribute by certain grainy facies such as leached Rudist Rudstone and Coated Grain Algal Floatstone presence in the vertical stratigraphic 4th order sequence.

Table A summarizes the key properties of the Reservoir ‘A’.

Table 1—Reservoir A key parameters

Property	Average Value	Range
Reservoir thickness	155ft	130 – 170ft
Permeability	20mD	0.01 to 3000mD
Porosity	23%	0-35%
Viscosity	0.22 cp	0.18-0.25 cp
Sor	15%	10 – 26%
Swi	34%	3 – 100%
RSI	1200 scf/bbl	800-1300 scf/bbl

Due to the diligent efforts of previous teams, a substantial dataset has been collected, including core samples from over 100 wells for reservoir rock properties calibration and geological analysis, alongside hundreds of pulsed neutron logs accumulated over the years. In addition, open hole logs are also available which enable proper evaluation of rock properties and fluid type.

Field Injection History

Since the 1970s, Field A has implemented water and gas injections strategies across various regions and reservoir sublayers for effective pressure maintenance (see [Fig. 2](#)). The initial phase involved peripheral water injection below the Free Water Level (FWL) and in the gas cap to mitigate gas cusping. Then, the field experienced a period of strategy inactivity (mothball). A few years later, lean gas was injected into the gas cap to sustain the field reservoir pressure. Subsequently, comprehensive water injection strategies were implemented within the oil pool for pressure maintenance as significant portion was being produced at this area. Concurrently, a series of gas injectors commenced operations in the northern sector of the field; these were later converted to water injectors. The most recent injection strategy has been deployed near the gas cap area, as part of the Up Dip, and Attic (UAD) development. [Fig. 3](#) illustrates the development areas and production/injection that took place at each development region. The development region is just an arbitrary boundary dividing the field into several sections. In the UAD, water injection operations serve two key purposes: maintaining reservoir pressure and creating a barrier to minimize hydrocarbon interchange between the oil rim and gas cap since the gas and oil are owned by different stakeholders. The most recent development phase involves simultaneous gas production alongside oil rim recovery. Initially, the reservoir's gas cap was rich in heavy components, contributing to a high condensate yield of approximately 100 stb/mmcsf. However, three decades of lean gas injection have altered this composition. Consequently,

accurately identifying the location of the remaining reservoir gas is essential for optimally positioning gas producers and maximizing condensate recovery

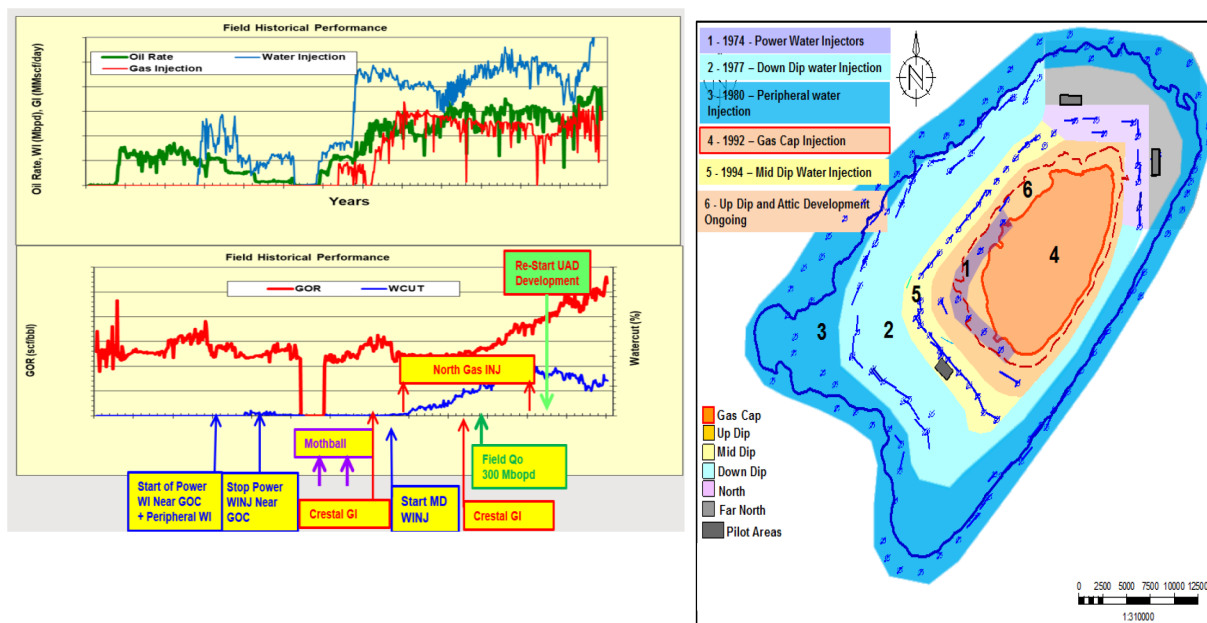


Figure 2—Field A injection history and the location

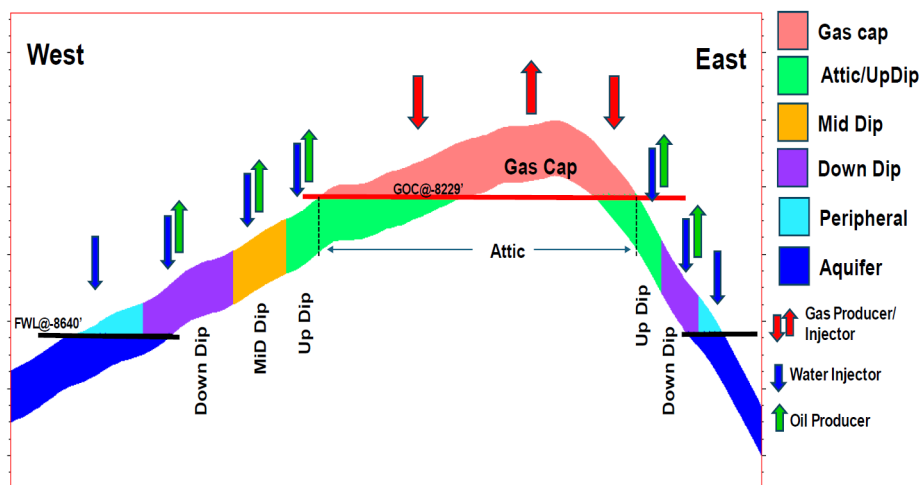


Figure 3—Cross section illustrating development area and injection production scheme

This paper examines fluid dynamics within the gas cap and Up Dip and Attic (UAD) region, which comprises approximately one third of the oil reserves in Reservoir ‘A’. The pronounced permeability contrast—ranging from 50 to 100 times—between the Upper and Lower zones significantly impacts flood front advancement. Timelapse PNL data reveal that injected water preferentially migrates above the oil, occupying the upper strata. Moreover, gas expansion aligns with stratigraphic layers rather than following the typical gas expansion behavior, subsequently migrating downward into the production zone due to pressure differentials associated with production offtake, creating gas fingering effect.

Data acquisition strategies

Data acquisition plays an important role in reducing project risk, particularly in Reservoir A due to the number of ongoing activities. As Reservoir ‘A’ represents the largest production source for the Field ‘A’, a

comprehensive Reservoir Management Plan (RMP) was developed and implemented to monitor reservoir conditions, maintain safety, and optimize recovery and economic output. The reservoir ‘A’ has accumulated a significant data set since the 1960s, including core, open hole logs, PVT analyses, PLT, pressure, Pulsed Neutron Logs (PNL), etc.

One of the main RMP objectives is to determine the fluid contact, fluid types and its associated saturations to monitor flood front advancement and evaluate injection efficiency. This was achieved using data from open hole logs, sampling and time-lapse pulsed neutron. Due to high average porosity and high formation water salinity, Pulsed Neutron Capture (PNC) with gas mode is suitable for Reservoir A. This paper will focus on time-lapse PNC evaluation.

Although an extensive data legacy is advantageous, it also presents certain challenges. The following outlines the primary concerns related to long-term PNC monitoring (Badruzzman & Fitz, Cased hole training material)*:

- Differences in tool technology/configuration with different service providers: spacing, crystal detector type etc*
- Differences in processing approaches *
- Differences in characterization/modelling approaches*
- Tool to tool inconsistencies
- Changes in borehole condition over time-lapse
- Differences in data quality acquired over time-lapse

Ideally, consistency in both tools and service provider would be maintained throughout the project. However, even within a single service provider, factors such as software updates, tool upgrades and interpreter could introduce variability (Fitz,2005). Moreover, relying exclusively on one service provider over a prolonged period may introduce bias and limit exposure to different technologies.

Going back to Reservoir ‘A’ main RMP objectives, fluid type and fluid contact are generally considered qualitative parameters and tend to be less impacted by variations in tool technology and service providers. Uncertainties in fluid contact primarily relate to depth accuracy, such as how well depths match between different time-lapse measurements. Fluid type is identified based on sigma properties and PNC ratios responses. A variation in sigma readings across time-lapses does not compromise the ability to differentiate between oil and water, as the sigma property exhibits a significant disparity between oil and saline water. However, while sigma may be less sensitive for distinguishing between oil and gas, differences in PNC ratios can aid this differentiation, making fluid type identification feasible. Now, the main challenge is to reconcile data from diverse service providers and technologies to derive accurate saturation changes over extended periods, thereby enabling effective evaluation of water and gas injection performance. These findings are integral to optimizing field development strategies, such as well spacing and the selection of injection/production layers.

In the past, the fluid type interpretation was typically oil/water/gas. However, it was first noted from near/far and near/long PNC ratios on the logs changes behavior within the gas zone itself (see Fig. 4). The shift in the ratios is notable and these responses could not be solved by using only one gas properties.

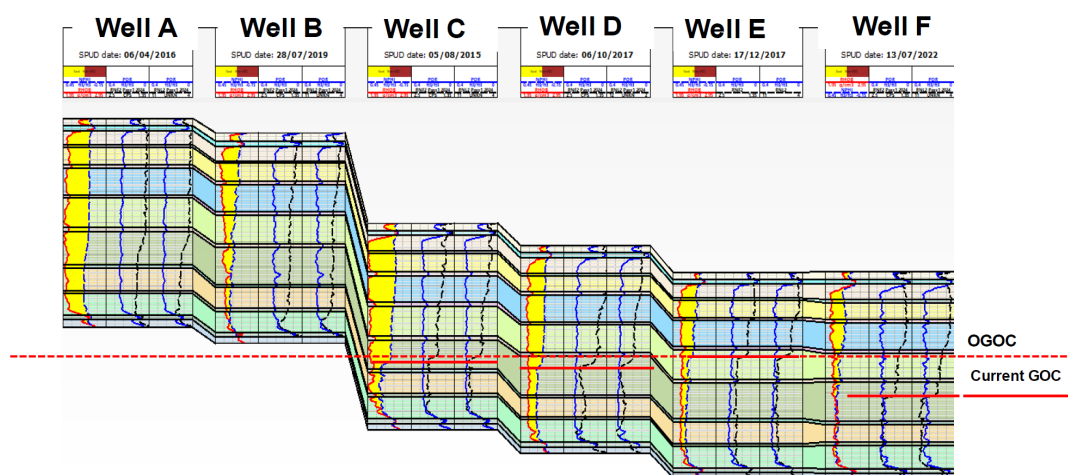


Figure 4—Neutron logs response in the gas cap (OH and PNC) arranged by TVDSS

To investigate this behavior, pressure data and PVT samples covering the vertical section were acquired. It was found that the shift on Pulsed Neutron Capture (PNC) logs responses is due to change in gas compositions within gas interval. Methane-rich injection gas remains at the top, while reservoir gas stays at the bottom, showing the system is not yet in equilibrium. Well E and F for example exhibit a clear step change between injection/reservoir gas as Well E and F are located 1.5 and 1km respectively away from gas injector with 25 years of gas injection history. The advancing front of injected gas is distinctly defined owing to the consistent supply of gas, and the gas has had limited time to diffuse. In contrast, wells located farther from injectors display less distinct contacts, suggesting that diffusion between injected and reservoir gases is more prominent. Nonetheless, the general distribution pattern—where injection gas occupies the upper zone and reservoir gas remains below—continues to hold true.

This response was observed in numerous wells throughout the gas cap and UAD regions, particularly in proximity to gas injectors. To address this issue, quad phase interpretations were implemented, providing the additional advantage of enabling the monitoring of injection gas movement.

Database

The primary factor guiding this timelapse interpretation workflow is the availability of data within the corporate database. To ensure process consistency, the workflow should integrate logs captured at each timelapse interval. Consequently, the procedure will be initiated with these specified data types. Table 2 summarizes the data available in Reservoir ‘A’, simplified only to include parameters related to PNC interpretation.

Table 2—Data availability

Category	Items	Availability
Logs	Open hole data	Raw and interpreted
	PNC data	Raw not available. Processed data such as sigma (formation, near, far, borehole, etc.), ratios, and count rates are consistently included in all acquired PNC datasets. Inelastic data is present in some datasets.
	Cement bond logs	Cement bond logs are available for all wells
PVT	Compositional analysis	PVT samples and analysis are routinely taken for water, oil, injection gas and reservoir gas. This information will be used to define fluid end points.

Category	Items	Availability
Core	-RCA/SCAL/MICP -Geological description	Data available for 110 wells with variable data quality.
Pressure data	Formation pressure	Extensive pressure data throughout field life are available

For open hole logs, core, PVT and pressure data, clean data sets which have been properly quality checked and evaluated are available.

Logs Interpretation

The PNC interpretation is conducted using a probabilistic module available in standard petrophysical software, ensuring the practicality of this workflow. Required input parameters include sigma, cased hole porosity, open hole porosity, and lithology. To minimize the possibility of crashing, as it is solving multiple fluids, the module is divided into two parts. Above the GOC, the module will solve for reservoir gas, injection gas and water while below the GOC, the model will solve for oil, formation water and injection water. Then, these two models will be recombined to provide the final interpretation.

Since Field ‘A’ is a multi-stacked reservoir, an oil producer or water injector located below Reservoir ‘A’ was used to acquire PNC. This approach provides broader areal PNC coverage across the field, supplementing the existing observers. However, to minimize uncertainties due to borehole conditions, only wells fitting below criteria were selected:

- Steel casing, single casing and tubing
- No perforation
- Fluid-filled borehole with known salinity
- Engaged borehole during open hole drilling – no big washout
- Good cement quality

This setup minimized uncertainties related to well completion or borehole fluid, allowing focus on the reservoir fluid.

For open hole data, lithology, porosity, and water saturation interpretations were calibrated against core data to ensure accuracy and reliability. The porosity uncertainty is $\pm 2\text{pu}$ while the water saturation uncertainties vary from ± 20 at the flank to $\pm 5\%$ at the crest.

The UAD area is situated approximately 300-400 feet above the free water level (FWL), where water saturation (S_w) remains at irreducible levels. In addition to S_w obtained from open-hole (OH) logs, the Saturation Height Function (SHF) in Reservoir ‘A’ is thoroughly calibrated using over 1000 MICP curves, providing reliable indication of original (pre-production) S_w conditions. If S_w measured from cased hole logging surpasses values from OH or SHF, it typically reflects the presence of injected water or mud filtrate that has not yet been displaced following drilling operations.

Rock and Fluid properties

The sigma for reservoir and injected gas were determined using Fitz and Badruzzaman (2021 PetroSkills cased hole training materials) using fluid compositional analysis in [Table 3](#). It could be seen that the injection gas is lean, mostly methane while the in-situ reservoir gas has more heavy components. This distinct difference is key in differentiating injection gas from reservoir gas.

Table 3—Typical Reservoir and injection gas properties

Components	Reservoir Gas	Injection Gas
	n_i	n_i
	(frac)	(frac)
CO ₂	0.0268	0.0354
C ₁ H ₄	0.5127	0.8971
C ₂ H ₆	0.1200	0.0563
C ₃ H ₈	0.1130	0.0073
iC ₄ H ₁₀	0.0330	0.0004
nC ₄ H ₁₀	0.0771	0.0005
iC ₅ H ₁₂	0.0289	0.0001
nC ₅ H ₁₂	0.0323	0.0002
C ₆ H ₁₄	0.0410	0.0002
N ₂	0.0031	0.0021
H ₂ S	0.0125	0.0002

The sigma matrix values for Limestone and Dolomite are sourced from the chartbook. Reservoir A is characterized as a clean limestone, and therefore minimal variation in its properties is anticipated. The original formation water has average salinity of 190,000 ppm while injected water has a corresponding average salinity of 250,000 ppm. The reservoir oil is light with an average API of 42 and viscosity of 0.22cp.

The fluid and rock parameters are summarized in Table 4. Density was not used but showed for comparison purposes. For hydrogen index (HI), the end points were determined using SNUPAR. The HI used in the model is corrected for temperature and salinity in the model. The water hydrogen index* is corrected for salinity and reservoir temperature/pressure in the probabilistic module.

Table 4—Average density and fluid parameters

Parameters	Density (g/cc)	Sigma (cu)	HI
Limestone	2.71	7.1	0
Dolomite	2.87	4.7	0.03
Oil	0.61	15	0.95
Formation Water	1.1	90	1 *
Injected Water	1.15	110	1 *
Injection gas	0.15	10.1	0.45
Reservoir gas	0.28	6.6	0.2

The sigma from Table 5 were used to recalculate synthetic bulk sigma values for various fluids using the equations below. This is used qualitatively to QC sigma log and fluid type determination.

Equations

$$\text{Syn_}\Sigma\text{_fluid} = \Sigma\text{_Lithology} + \Sigma\text{_fluid} \quad \text{Equation 1}$$

$$\Sigma\text{_Lithology} = V\text{Calcite} * \Sigma\text{Calcite} + V\text{Dolomite} * \Sigma\text{Dolomite} \text{ where } V\text{Calcite} + V\text{Dolomite} \quad \text{Equation 2}$$

$$\Sigma\text{_fluid} = S\text{wirr} * \Sigma\text{Water} + (1 - S\text{wirr}) * \Sigma\text{ReservoirGas/InjectionGas/Oil} \quad \text{Equation 3}$$

Irreducible water, determined from open hole logs or through the SHF, was integrated into the above equation to enable comparison between log-derived results and bulk sigma data provided by service providers. Normalization is then applied at dense interval (dense above/below reservoir and intra-dense), interval of known fluid (unswept oil zones) and water zones (if close to FWL).

Generally, the service provider sigma (corrected for borehole and diffusion) match very well with the above equations within ± 2 capture units. If the difference between sigma synthetic, previous time lapse, and nearby well measurements is significant, the data is flagged as yellow or red, indicating that the resulting interpretations involve high uncertainty. In such cases, fluid contacts and fluid type may be used for modeling, but saturations should not be included. Typically, these discrepancies can be traced to a specific tool generation and period.

Cased hole porosity

Cased hole (CH) neutron porosity poses the highest uncertainties in this workflow. The biggest issue is that the CH neutron porosity is inconsistent throughout time-lapses. After processing, some service providers proceed to further normalize cased hole porosity values to align with OH neutron readings. While this approach can facilitate reconciliation between OH neutron and PNC data, certain interpreters may excessively adjust the data to match open hole neutron results, without considering fluid changes that occur post-drilling. This underscores the importance of thorough data quality control by both the service provider and the company. Although such discrepancies are corrected, in a large reservoir, where hundreds of PNC logs are acquired annually, it is possible for several wells to go unnoticed. Consequently, maintaining thorough and vigilant supervision is essential to avoid these errors.

Sullivan, Michael & Daving Belanger, (2012) mentioned that PNC ratios, once rescaled, matches with OH neutron porosity very well. Similar observation was observed in Reservoir A. Therefore, the time-lapse ratios could be rescaled to match open hole neutron log and any reduction from this baseline is indicative of gas responses. However, open hole neutron porosity is highly sensitive to drilling conditions and borehole corrections. Neutron porosity measurements obtained via wireline logging and Logging While Tripping (LWT) are subject to greater invasion effects compared to those acquired under Logging While Drilling (LWD). In addition, the neutron porosity corrections also differ from one service provider to the other.

Well G and H in Fig. 5 were logged using LWT as technology trial back in 2019/2020. Density-Neutron separation could be seen from OH logs but not as prominent in Fig. 4 due to invasion caused by high mud overbalance. The gas saturation calculated from OH LWT will be suppressed while GOC determination is subjected to high uncertainty. In this instance, the data from open hole logs would not be used as it is subjected to high uncertainty and priority will be given to the PNC data.

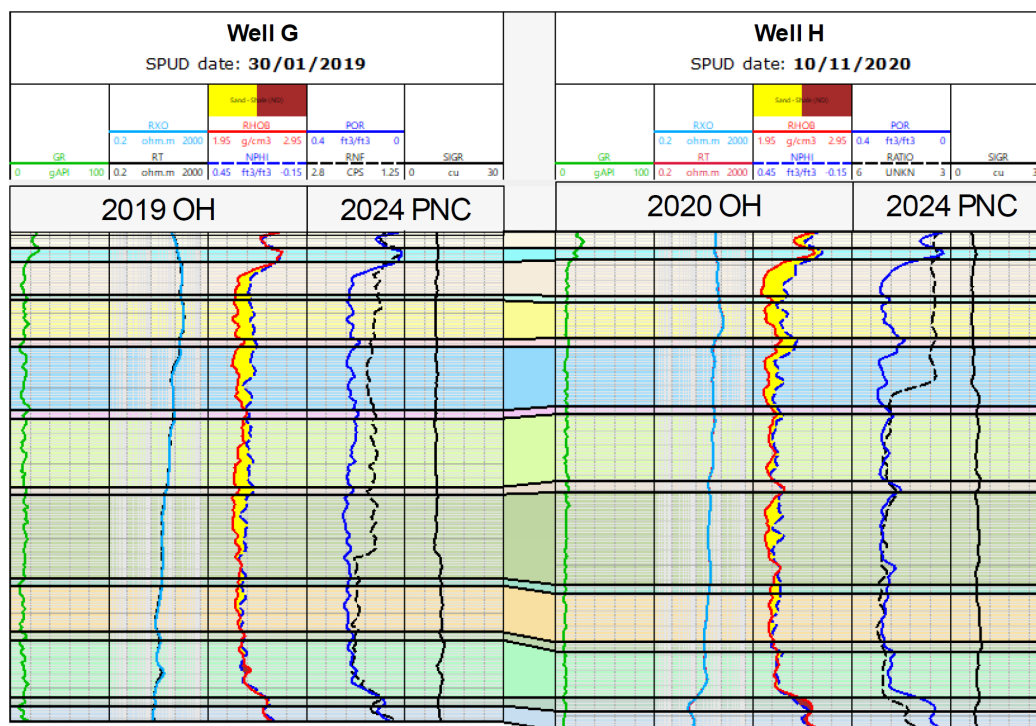


Figure 5—Comparison between open hole LWT and PNC - Gas flow back post drilling

Therefore, instead of using open hole neutron porosity, open hole porosity which has been calibrated to core data, serves as the most practical and reliable baseline. In this workflow, the near/far ratio from three detector PNC tools were used because it has the deepest depth of investigation and is available in time-lapse datasets in the UAD area. As all data are adjusted relative to a recognized baseline, the error will be systematic rather than random. Like sigma, the control points for rescaling will be at dense interval (dense above/below Reservoir 'A' and intra-dense), interval of known fluid (unswept oil zones) and water zones (if close to FWL). The resultant porosity rescaled from near/far ratio will be called cased hole porosity. For fluids such as oil or water, the cased hole porosity corresponds to open hole porosity. Any reduction observed in cased hole porosity indicates the presence of gas, which is then further characterized to differentiate between injection and reservoir gas.

The rescaling in Reservoir 'A' was accomplished effectively as a result of the following factors:

- Open hole porosity calculations are robust and well calibrated to core data (Fig. 6).
- Dense interval: Thick dense above and below Reservoir A together with distinct intra-dense within the reservoir layer (Fig. 6)
- Unswept oil: because of high permeability streaks, the oil in Lower zones remain unswept which is around 80-100ft thick. This provides good control for rescaling (Fig. 7)

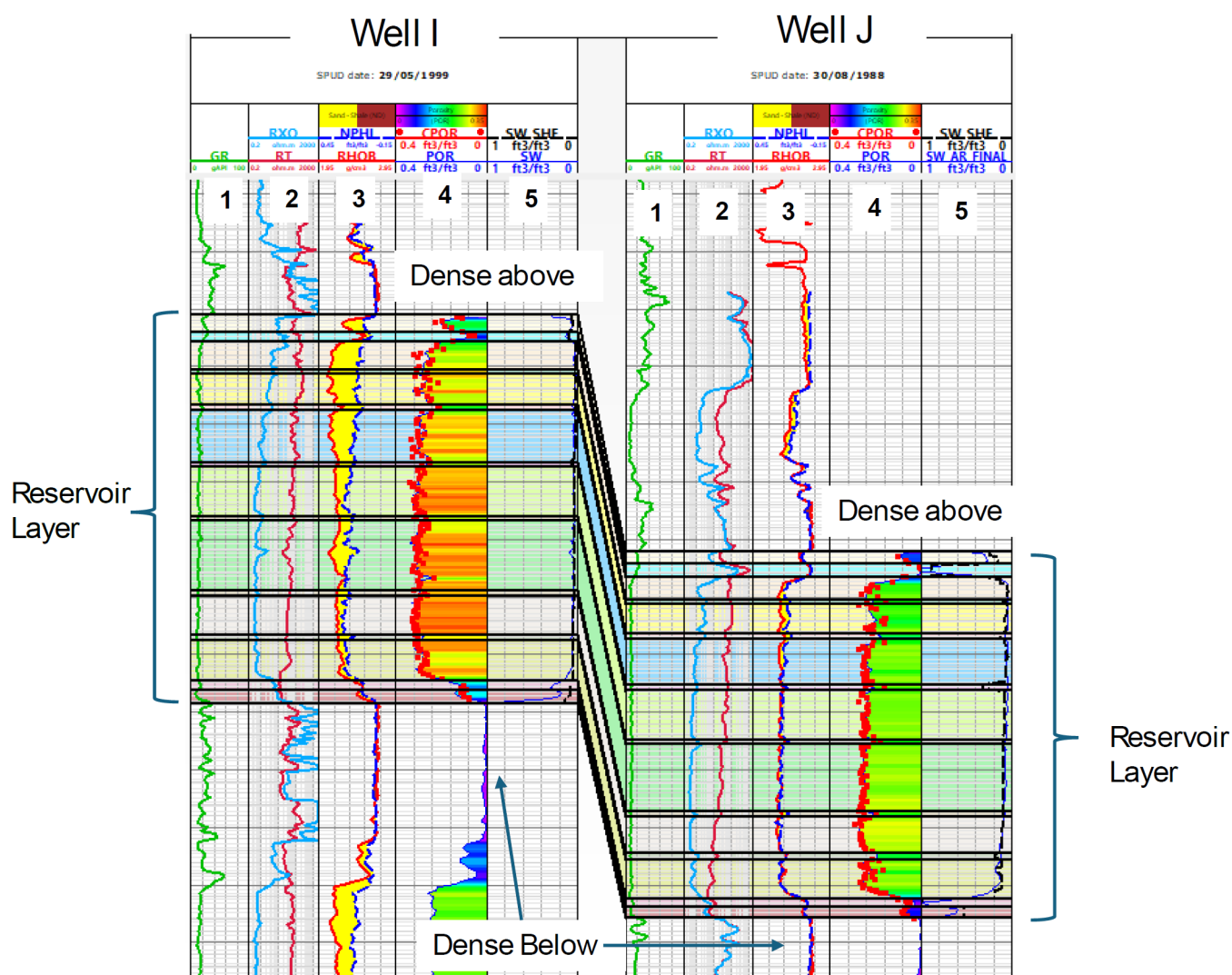


Figure 6—Open hole logs responses showing dense above and below Reservoir A .
T1:GR, T2:RT/RXO, T3:DEN/RHOB, T4 : Open hope porosity/Core porosity, T5:SW/SW_SHF

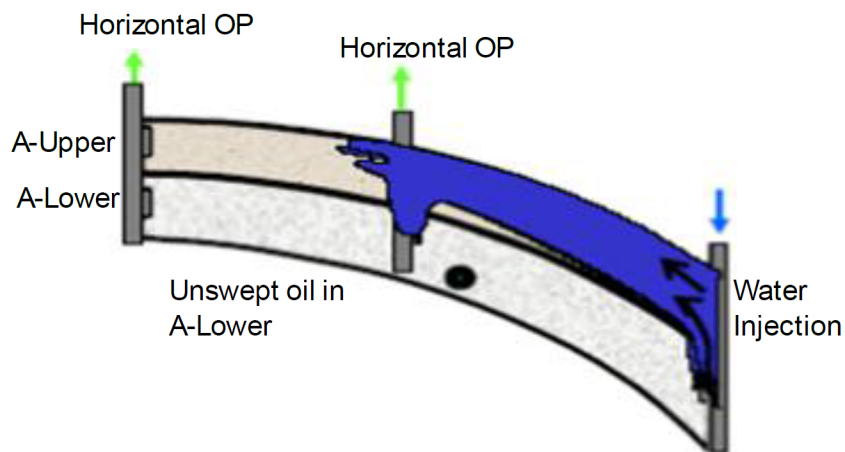


Figure 7—Water gravity override in Reservoir A due to High permeability streaks leaving unswept oil in A-Lower

The workflow is summarized below:

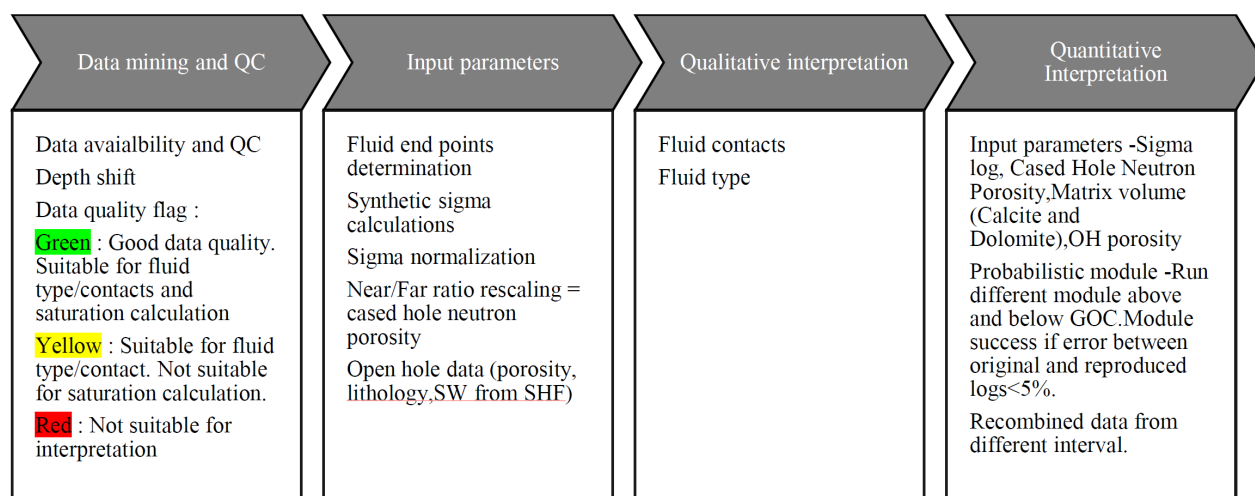


Figure 8—PNC Interpretation workflow

Results

Case Study 1

Well E is an observer in Reservoir 'A' that was drilled in 2017. The well is in Eastern Flank, 1.5km away from Lower zone active gas injector that has been injecting cumulatively 484 BSCF since 1999. In this well, pressure data was taken so the fluid gradient could be determined (Track 5, Fig. 9). From pressure data in Fig. 10, three distinct fluid density could be observed where at the top of reservoir has gas with density of 0.16 g/cc (red points) followed by gas with density of 0.27g/cc (orange points) which could be related back to injection and in-situ reservoir gas respectively. Slight log differences from open hole neutron in the gas interval were observed but not enough to differentiate the fluid properties. Below the gas, the oil points in green formed a very uniform gradient of 0.59g/cc. In 2019, PNC were acquired in this well. Near/Far and Near/Long ratio showed distinct changes in the gas zone corresponding well to fluid gradient changes observed from pressure data. Using the above workflow, PNC interpretation showed injected gas at the top, followed by reservoir gas and oil. No water saturation changes were observed and expected as there is no nearby water injector.

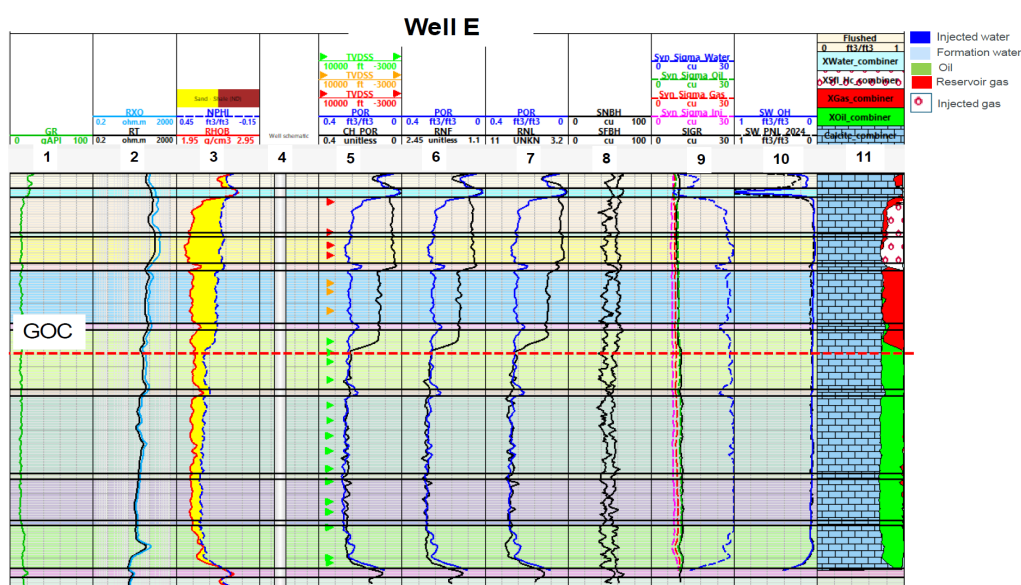


Figure 9—PNC well interpretation in Well E

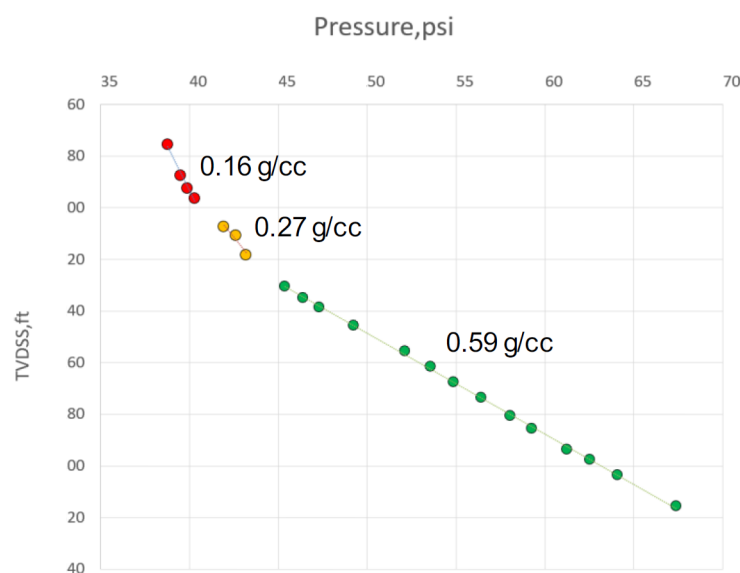


Figure 10—Well E Pressure plot, the color code corresponds to point in Track 5

Case Study 2

By utilizing the above workflow, the injection gas plume could be detected areally and vertically. Fig. 11 showed the variation of injection gas vs in-situ reservoir gas at different locations of the field. The correlation is in TVDSS. The location of wells is depicted in Fig. 12. Well K, which is located closest to gas injector saw injection gas down to Lower zones meaning most of original gas cap were already displaced at this location. Well L was located further away from the gas injector, saw injection gas in Upper zones. While at Well M location, injection gas has not arrived in 2019 indicating the area with the highest condensate recovery. In Well L, CGR from PVT samples were acquired and used as validation for the proposed workflow. This will be discussed more under condensate tracking and gas development.

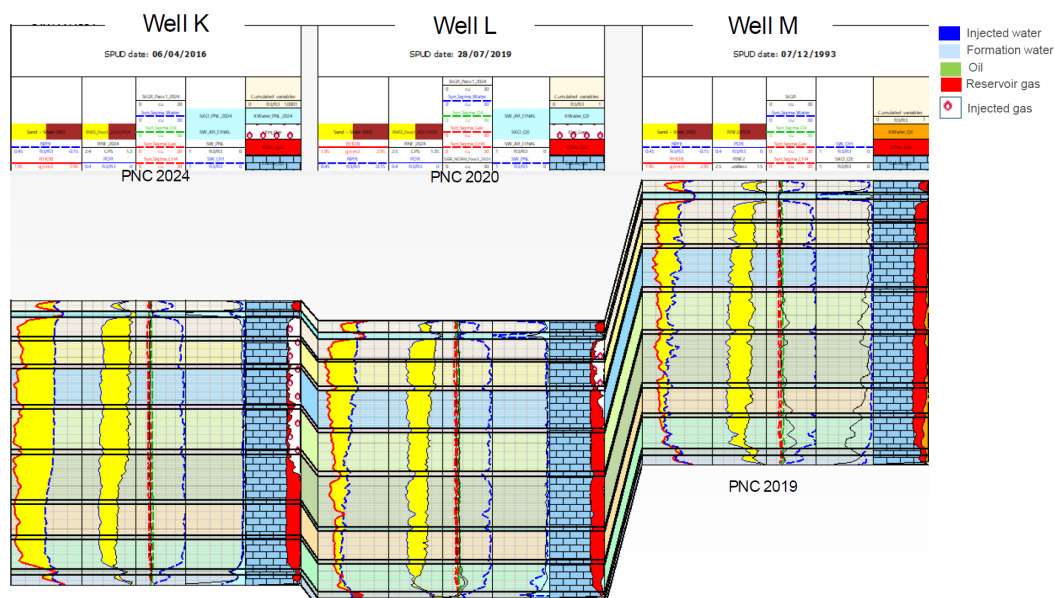


Figure 11—PNC in gas cap-injection gas variation depending on well location

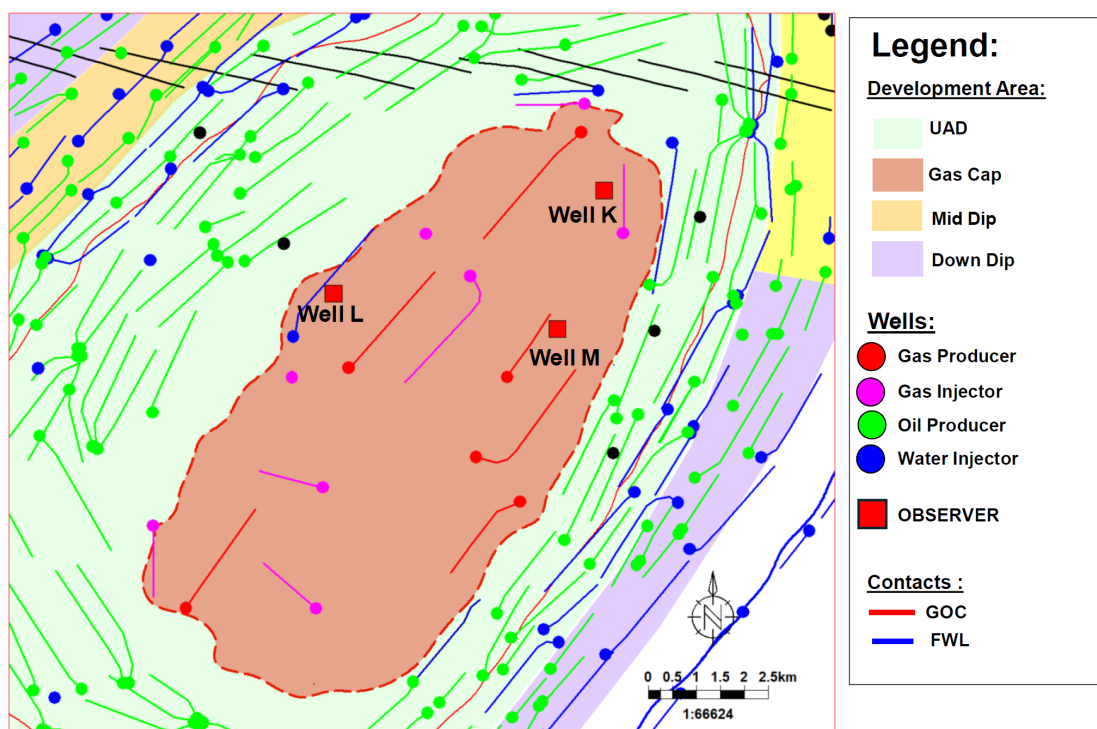


Figure 12—Location of K,L,M and the nearby gas injectors.

Case Study 3

Well N is located in the western part of Field A and features multiple time-lapse PNC measurements from 2016 to 2025 as depicted in Fig. 13. The lithology track was removed to showcase the fluid movement more clearly.

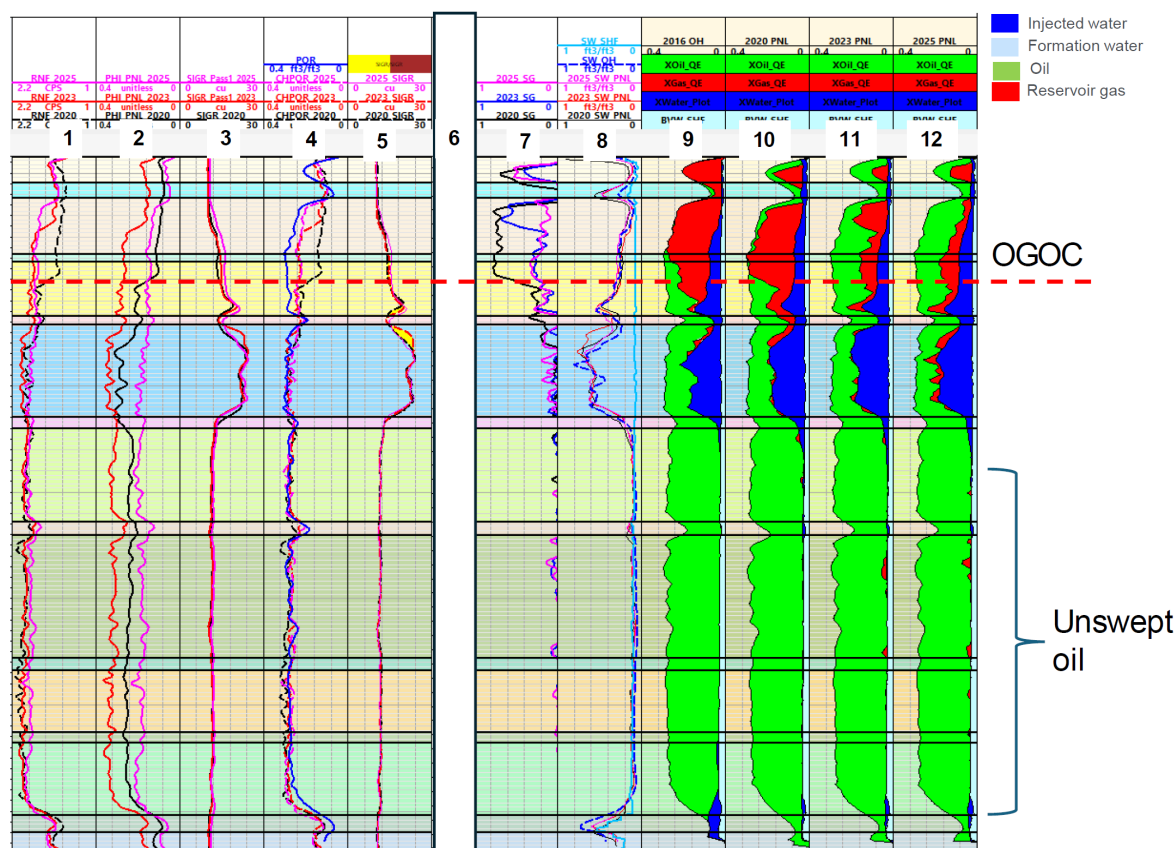


Figure 13—Time-lapse interpretation for Well N

Track 1 represents the near/far ratio, Track 2 displays the PNC porosity and Track 3 showed the sigma values provided directly by the contractor. While the trends across the time-lapse data are comparable, there are notable discrepancies in the absolute values, particularly with respect to PNC porosity. The use of these inconsistent values will result in ambiguity during interpretation. Track 4 displays the cased hole neutron porosity calculated from the rescaled near/far ratio while Track 5 showed the normalized sigma. Track 7 and 8 present the gas and water saturation over time, while tracks 9 to 12 illustrate how the fluid flood front evolves.

In 2016, when the well was drilled, it was observed that gas had moved below the original gas oil contact (OGOC). This well is situated at a distance from any gas injector, with the closest being 4.5 km away; therefore, the detected gas is attributed to migration from the in-situ reservoir gas. The gas travelled down following high permeability streaks in the Upper zone to compensate for production offtake nearby Well I. There are two water injectors near Well N: one updip (700m away, active in the 70's and has been decommissioned after 4 years of injection) and one downdip (450m away, active since 2015). Similar to gas, the injected water finds its way through the Upper zone through high-permeability streaks. It could be seen that the water observed in Track 9 to 12 that the water saturation remains relatively constant from 2016-2025 – most likely injected water from the decommissioned water injector. For the gas, significant decrease was seen throughout the years - the downdip water injector has mobilized the oil bank into the gas cap, decreasing the gas content.

Case Study 4

Well O is positioned downdip relative to Well N where the original fluid is oil as depicted in Track 9, Fig. 14. Fig. 14 presents the PNC log responses collected from 2014 to 2024. Tracks 1, 2, and 3 display the RNF, PNC porosity, and sigma data as provided by the contractor. Tracks 4 and 5 represent the rescaled porosity

Systematic spatial enhancement of the High Permeability Streaks (HPS) using dynamic feedback from time-lapse PNL data significantly improved the predictive capability of the full-field model. This workflow resulted in a measurable improvement in history match quality, where the water cut match improved from 53% to 86%, and the GOR match improved from 43% to 75%. These improvements highlight the robustness of integrating dynamic surveillance data into static model updates, ensuring that the reservoir behavior is better captured and future performance predictions are more reliable.

A key outcome of this exercise was a fundamental shift in the geological understanding of the HPS concept. The initial interpretation assumed that the magnitude of permeability within streaks was the dominant control on flood front propagation. However, detailed PNL-versus-model saturation analysis revealed that it is not solely the absolute magnitude of permeability, but rather the contrast between the HPS and surrounding matrix that governs fluid movement. This contrast creates preferential flow pathways that accelerate localized flood advancement through the streaks, subsequently leading to broader displacement and flooding of the Lower zone permeability matrix.

To reflect this revised understanding, the matrix rock type description was improved from Rock Type A to Rock Type B, both of which represent geologically consistent coated grainstones but with varying degrees of sorting and cementation. Fig. 15 illustrates the Vertical Proportion Curve (VPC) and porosity permeability cross-plot pre and post model integration. This subtle yet critical refinement preserved the volumetric balance of the reservoir, as the overall oil in place was maintained within $\pm 5\%$, owing to the similarity in saturation-height functions between the two rock types. However, the key difference lies in the variation of entry pressures, which better captured the timing and pattern of water and gas breakthrough observed in the field.

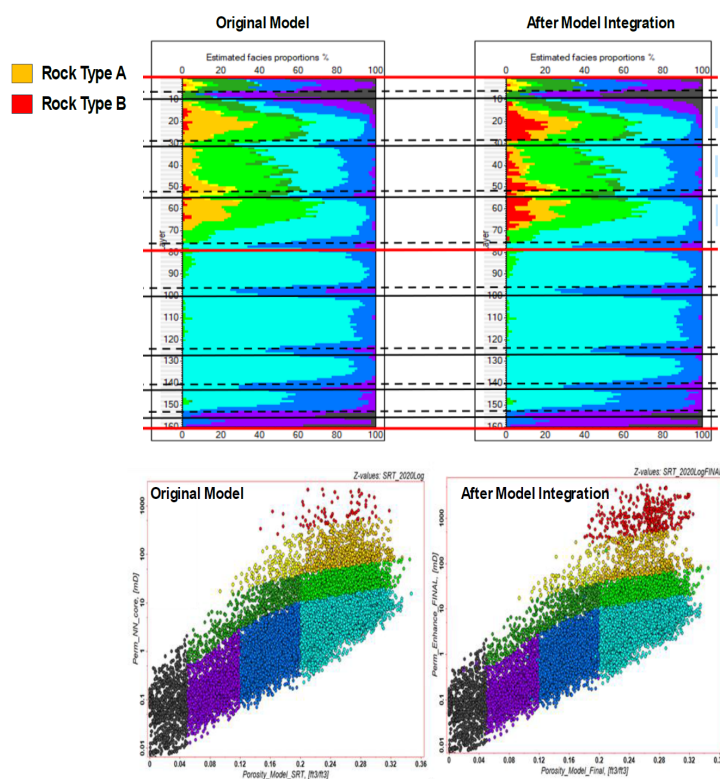


Figure 15—Improvement in Rock Typing post PNL integration

By integrating this enhanced geological concept with dynamic calibration, the reservoir model evolved from being a static representation to a predictive decision-making tool. This not only provided greater confidence in ongoing waterflood and gas management strategies but also established a clear workflow for

future model updates that link surveillance data with geological and petrophysical characterization. More details on this workflow are described by [Khan et. al \(2021\)](#).

Condensate tracking and Gas Development. The significant volume of gas injected throughout the years has significantly changed the gas cap composition. Therefore, to track the injection gas movement, Dual Component Equation of State, Dual EOS was carried out in Reservoir A with the primary objectives to study how the Condensate Gas ratio (CGR) changes because of continuous gas injection. In this field, the target is to produce the in-situ reservoir gas which is rich in condensate, the injection gas is lean and has minimal CGR recovery. More details on Dual EOS simulations are available in [Wang et.al \(2020\)](#).

[Fig. 16](#) shows the CGR maps for Reservoir ‘A’ gas cap. The map on the center, for Full Field Model (FFM) layer 5, 20 and 30 (out of a total of 54 layers) is from for VIP FFM initialization run. The scale in the legend is for CGR in stb/mmscf, with the maximum scale at 150 stb/mmscf.

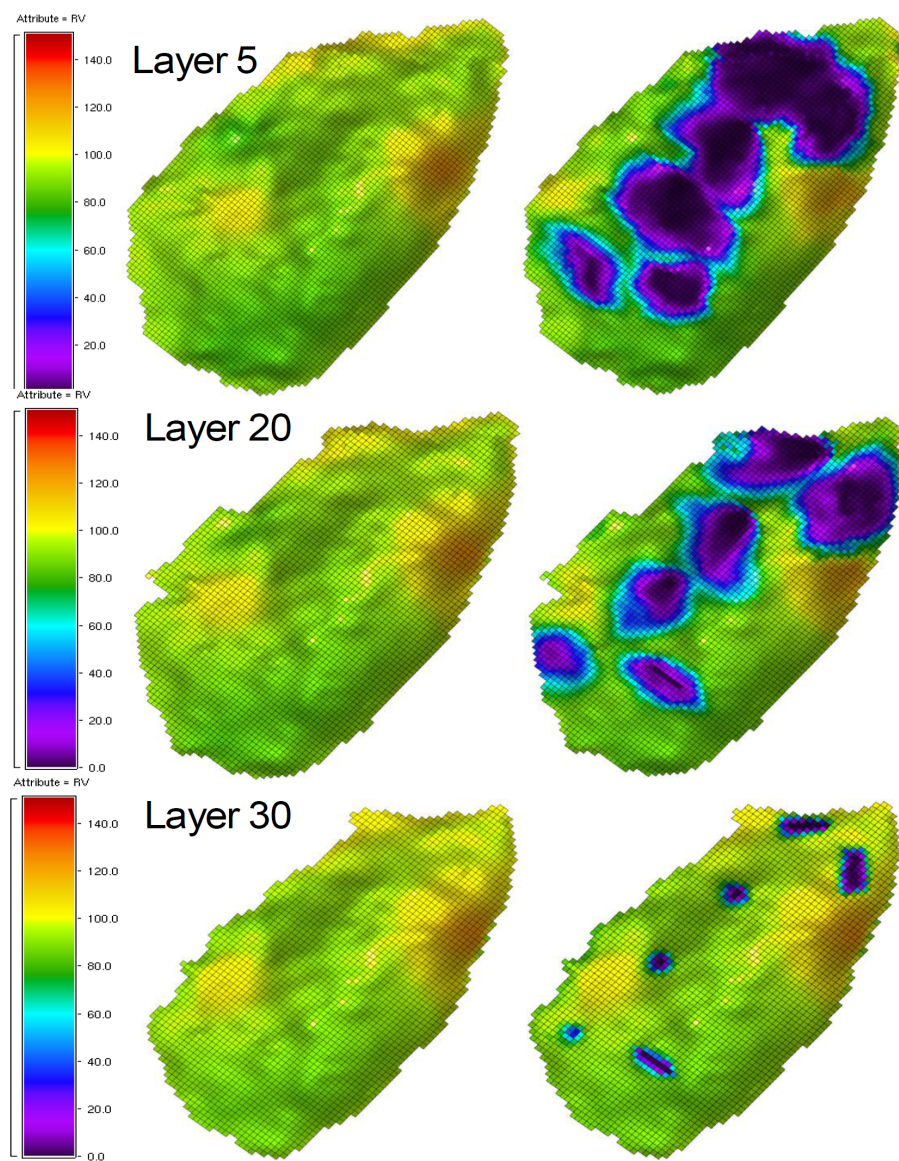


Figure 16—Reservoir A gas cap CGR at different layers across Reservoir A – initialization and end of history match

At layer 5, the plume of gas injection is visible and wide-spread, though quantitatively it may be affected by numerical dispersion due to the large cell size in FFM. It is obvious that at layer 20, the lean gas plume is, though still wide-spread, visibly smaller in footprint. At layer 30, the injection lean gas plume is only limited

to the immediate vicinity of gas injectors. This is a clear indication that the impact of lean gas injection is felt primarily in the crestal area of the gas cap, due to the density contrast between injection gas and fairly rich in-situ gas cap gas. The impact of lean gas injection on the CGR change in the gas cap can be readily seen from Fig. 17 where the CGR statistical distributions are compared. The average CGR is reduced from 93.4 stb/mmscf at initialization to 76.9 stb/mmscf at the end of the history match run.

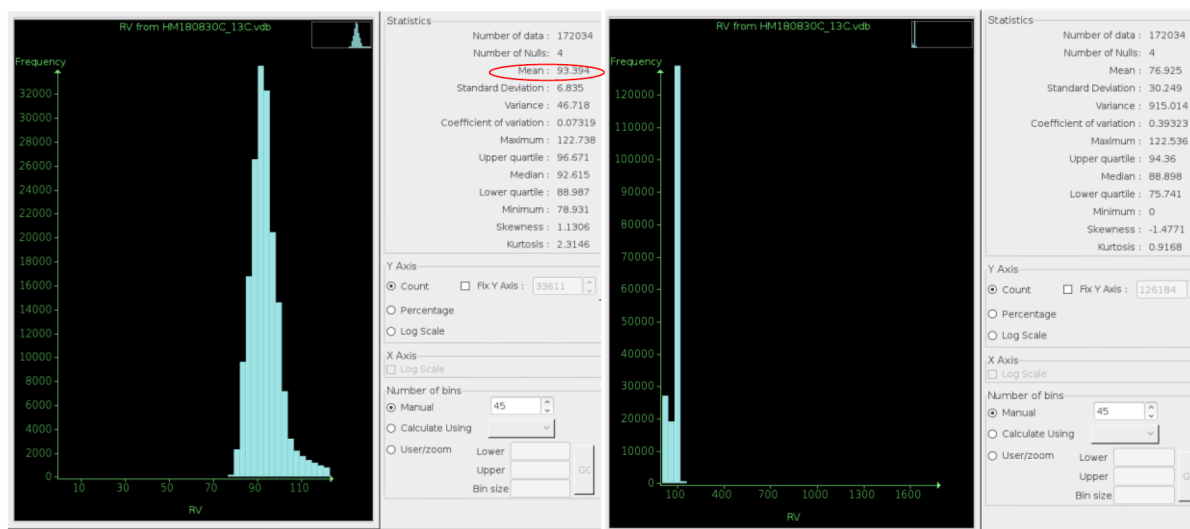


Figure 17—KH-2 gas cap CGR change – initialization and end of history match

Well L was drilled as an observer in 2019. As part of the Dual EOS model validation, PVT samples were collected along the vertical section in Well Gas, as shown in Fig. 18. These samples underwent analysis to determine the CGR values and to assess CGR variation with depth. In 2020, baseline PNC logs were acquired. Tracks 1 to 4 display the PNC log responses, while Track 5 presents the CGR from the Dual EOS model (2020, black line) alongside the actual CGR derived from PVT analysis (red dots). Track 6 illustrates the interpreted results using the proposed workflow, identifying injection gas in the upper reservoir and in-situ reservoir gas in the lower interval. Interpretation of the PNC logs aligns closely with the CGR trends observed in the PVT data. Additionally, the modeled CGR demonstrates a similar trend, supporting the calibration of the Dual EOS model. The proposed workflow can be applied to all PNC data, enabling more wells as calibration points and increasing model confidence with additional control points.

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